

Nikita Agrawal

AI Engineer

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Professional Summary

AI Engineer with **6 years** of software engineering experience, including building production Generative AI applications. Experienced in Python, LLMs, RAG, Agentic AI, Prompt Engineering, AI evaluation, enterprise API integrations, and cloud AI services. Built production AI systems using Azure OpenAI, FastAPI, FAISS, Pinecone, and enterprise APIs. Experienced with Cursor and Claude Code for AI-assisted development, agentic workflows, debugging, code review, and rapid prototyping.

Experience

All Known Services / Nobelistech Services LLP

AI Engineer (Contractual)

Aug 2026 – Sept 2026

- Supporting AI, data, and analytics initiatives across client engagements, including initiatives associated with Cleartelligence and DoorDash.
- Building and scaling AI-powered tools, workflows, and applied AI solutions as part of company AI initiatives for business and client use cases.
- Collaborating with stakeholders to identify and implement high-impact AI use cases and support client initiatives.
- Working on Agentic AI workflows for accounting use cases, using Cursor and Claude Code for AI-assisted development, agent/workflow implementation, prompt iteration, debugging, and rapid prototyping.

Matilda Cloud

Machine Learning Engineer

Jan 2026 - Aug 2026

- Built production LLM-powered cloud recommendation systems using Python across AWS, Azure, and OCI, including AI/LLM workflows for recommendation validation, prompt engineering, and workflow automation.
- Developed FastAPI-based AI services integrating cloud APIs, SQL databases, pricing APIs, and backend data pipelines.
- Used Cursor, Claude Code, NVIDIA CUDA-enabled PyTorch, and GPU acceleration for AI development, experimentation, inference, debugging, and LLM evaluation.

Tata Consultancy Services

System Engineer - AI Engineer

Mar 2021 - Jan 2026

- Designed and developed an enterprise RAG-based knowledge chatbot integrating Jira, Confluence, and SharePoint using Azure OpenAI, Azure AI Search, FAISS, Python, embeddings, and FastAPI.
- Integrated enterprise REST APIs, SQL databases, authentication, and backend pipelines, while working on prompt engineering, AI evaluation, retrieval optimization, Computer Vision, and Deep Learning using Python, OpenCV, TensorFlow, and PyTorch.

Overseas Technologies

Full Stack Developer

Aug 2020 - Feb 2021

- Developed Android and web applications using Java, Python, Kotlin, JavaScript, PHP, and Firebase.

Nikist

Content Writer

Aug 2018 - Nov 2019

- Created engaging and informative content through technical writing and e-books for websites and blogs.
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Education

Master of Technology in AI and ML
Bachelor's in Information Technology

Rashtriya Raksha University - 2024
Gujarat Technological University - 2020

Technical Skills

- **Languages:** Python, Java, SQL, C++
 - **AI / GenAI:** LLMs, Generative AI, Agentic AI, AI Agents, Prompt Engineering, RAG, AI Evaluation, Function Calling, AI Workflow Orchestration
 - **ML / Computer Vision:** Machine Learning, Deep Learning, OpenCV, CNNs, Computer Vision, OCR, Object Detection, Transfer Learning
 - **Frameworks:** PyTorch, TensorFlow, LangChain, Hugging Face, FastAPI, Scikit-learn
 - **Cloud & AI:** Azure OpenAI, Azure AI Search, AWS, GCP (Vertex AI, BigQuery), OCI
 - **Databases / Vector Search:** PostgreSQL, MySQL, FAISS, Pinecone, OCI Autonomous Database
 - **Backend & APIs:** REST APIs, Enterprise API Integration, Backend Data Pipelines, SQL, Workflow Automation
 - **AI Development:** Cursor, Claude Code, GitHub Copilot, AI-assisted Development, Prompt Iteration, Debugging
 - **DevOps & GPU:** Docker, Git, CI/CD, NVIDIA CUDA, CUDA-enabled PyTorch
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Projects

AI Initiatives & Agentic AI Solutions (All Known Services / Nobelistech Services LLP)

- Contributing to AI initiatives and applied AI solutions supporting business and client use cases, including engagements associated with Cleartelligence and DoorDash.
- Building AI-powered and Agentic AI workflows for business automation, including accounting-focused use cases.
- Using Cursor and Claude Code for agent/workflow development, prompt iteration, debugging, and rapid prototyping.

Technologies: Python, LLMs, Generative AI, Agentic AI, AI Agents, Prompt Engineering, Cursor, Claude Code, REST APIs, SQL, AI Workflow Orchestration

Cloud Rightsizing & Pricing Recommendation Agent (Matilda Cloud)

- Developed AI-driven cloud rightsizing recommendation agents for AWS, Azure, and OCI, combining utilization metrics, pricing APIs, SQL metadata, and LLM reasoning.
- Built recommendation workflows for Azure VMs, AWS DynamoDB, and OCI Autonomous Database covering resource utilization, pricing, and performance optimization.
- Integrated cloud APIs and backend data pipelines and monitored AI response quality, latency, and prompt behavior in production.

Technologies: Python, Azure, AWS, OCI, FastAPI, LLMs, REST APIs, SQL, Cloud APIs, Cursor

Data Assessment Conversational Bot (Matilda Cloud)

- Built AI agents for assessment summarization, graph/chart visualization, and natural-language insights from cloud assessment data.
- Implemented RAG agents using FAISS/Pinecone and evaluated retrieval quality using RAGAS and G-Eval against gold-standard datasets.
- Monitored AI application performance, latency, response quality, prompt behavior, and observability metrics in production.

Technologies: Python, LangChain, LLMs, RAG, FAISS, Pinecone, FastAPI, RAGAS, G-Eval, AWS, Azure, GitHub Copilot, REST APIs, SQL, Claude Code, Backend Pipelines, Regression testing

Financial Innovation LLM System through Prompts (TCS)

- Built an LLM-based financial assistant using prompt engineering and custom prompt pipelines for financial insights and analytics.
- Improved response quality through structured prompt templates and iterative prompt refinement.

Technologies: GPT-4, Python, FastAPI, GCP BigQuery, FAISS, Pinecone, Scikit-learn

Cache Cleanup and Data Management AI System (TCS)

- Developed an ML-based cache optimization system that reduced manual effort by 65% and automated stale-data detection.

Technologies: Python, AWS, FastAPI, PostgreSQL, Docker, CI/CD

Chatbot Knowledge System (TCS)

- Built an enterprise RAG-based knowledge chatbot integrating Jira, Confluence, and SharePoint, with prompt engineering and retrieval workflows.

Technologies: Python, LLMs, Azure OpenAI, Azure AI Search, FAISS, FastAPI, Hugging Face, REST APIs, SQL, Claude Code, Backend Pipelines, NVIDIA CUDA/PyTorch

Telecommunication Crime Investigation through Speech Recognition Technology

- Designed a ML-based voice pattern recognition system for law enforcement to detect suspicious voice communications.

Technologies: AI-SR, ESR, Acoustic Modeling, NLU, Speech Segmentation, NLP

Edge Impulse AI Sensors Anomaly Detection

- Developed an AI-driven anomaly detection system for IoT sensors using ESP32 microcontrollers.

Technologies: ESP32, Edge Impulse AI, IoT Sensors, Weather Toolkit, BiLSTM, ApacheNLP

Image Analysis & Detection System

- Developed a computer vision pipeline for image preprocessing, classification, and object detection using CNNs and transfer learning, with model optimization through hyperparameter tuning.

Technologies: Python, OpenCV, TensorFlow, PyTorch, CNN, Transfer Learning, Object Detection, Image Classification

Transport Domain (TCS) - QA and API validation using PostgreSQL.

Vegrotte App - Android app with Firebase integration.

Stress Buster - Security-focused vulnerability integration app.

Summit Gate - Conference Management System

Smart Parking System - IoT-based parking interface.

Publications

AI-Equipped Edge-IoT Devices in Security - *Taylor and Francis LLP*

Feb 2024

Stress Buster - *IJRASET*

Apr 2020

Prevention of Fraud Transactions Using Fingerprint Verification - *JETIR*

Feb 2019

Certifications

- Data Analytics & Insights - Advanced NLP with Python for ML
- Artificial Intelligence Executive Briefing (Udemy)
- Advanced AI Transformers for Computer Vision
- Foundational Generative AI
- AI, Machine Learning & Deep Learning Workshop (GTU)
- ICSI | CNSS Certified Network Security Specialist
- Shape AI Bootcamp - Python & Cyber Security, Web Development
- LinkedIn Learning - Writing Emails, Interpersonal Communication, Lean 6 Sigma, Remote Work Foundation